

OSPF

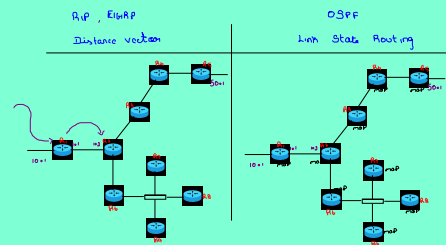
open Shortest Path First

Link State Routing Protocol

metric = cost

It contains 3 tables

- * Neighbors Next Hop
- * Link State Database Map
- * Routing Best Route



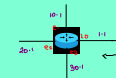
- Router ID
- Area ID
- LSDB

Router ID

x.x.x.x

1=Out 1

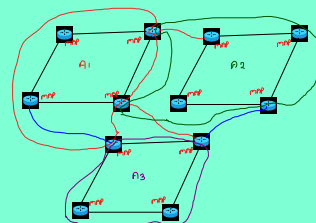
- * manually assign
- * highest IP in loopback int
- * highest IP in physical int



Area ID

0.0.0.0

OSPF



Web

wild card bits
→ matching bits

0 - core

1 - don't care

OSPF Neighbor Ship

Hello 224.0.0.5
→ OSPF multicast IP

Same n/w
Same Area ID
unique Router ID
Same time

OSPF time

Hello time = 10 Sec

Dead time = 4 * 10 = 40 Sec



OSPF metric :- Cost

$$= \frac{10^8}{\text{bandwidth in bits}}$$

$$= \frac{10^8}{10,000 \times 1000}$$

$$= 10$$

RIP - 120

EIGRP - 90

OSPF - 110

Static - 1

Default - 0